Errata

On Coppock, Hill and Vavreck (2020), *Science Advances* 6(36)

The maintained replication code behind these errata is at https://github.com/active-maintenance-aec/coppock_hill_vavreck_2020 and is archived at Zenodo at https://doi.org/10.5281/zenodo.24168360, a DOI that always resolves to the latest version.

This note lists nine errors in "The small effects of political advertising are small regardless of context, message, sender, or receiver: Evidence from 59 real-time randomized experiments," published in *Science Advances* 6(36), doi 10.1126/sciadv.abc4046. **None of them changes a conclusion of the paper.** That is an adjudication rather than an assurance: each entry below was taken back to the sentence a referee would have said rests on it, and each note says what the corrected value is now the warrant for. The first eight are recorded because the numbers themselves are wrong and a reader who quoted them would be misled; the ninth is a dropped letter, recorded on the same principle.

Items 1, 2, 3, and 7 are in the main text, items 4, 5, and 8 in section A of the supplementary materials, and items 6 and 9 in Materials and Methods. Every corrected value below is computed from the deposited replication archive (10.7910/DVN/TN7KWR) at the time this document is rendered. The maintained code is at github.com/active-maintenance-aec/coppock_hill_vavreck_2020.

1. The CATE homogeneity p-value for vote choice is misstated by a factor of ten

Published, Results:

> For the set of conditional average treatment effects (CATEs), P values are 0.0002 and **0.96**.

Corrected:

> For the set of conditional average treatment effects (CATEs), P values are 0.0002 and **0.0961**.

Table S2 of the same article reports 0.0961 for this test, which is what the deposited code produces. The test still fails to reject the null of homogeneity at the 5 per cent level, so the sentence two paragraphs later, that homogeneity is not rejected in three of four opportunities, stands as published.

What the correction changes is how much weight that sentence can carry. The three tests the article does not reject now run 0.070, 0.092, 0.096, so all three sit between 0.05 and 0.10 and the count would be none of four at the 10 per cent level. The published 0.96 was the only one of the four

p-values no conventional threshold could touch. The passage does not rest on the tests: it says so itself, declining to affirm the null and relying instead on the random-effects estimate of the true standard deviation of the effects, which for these CATEs is 2.3 percentage points on vote choice and is unchanged.

2. The general election coefficient disagrees with Table 1

Published, Results:

> On average, general election advertisements move candidate favorability by **0.121 scale points** more than primary advertisements, although the estimate is not precise enough to achieve statistical significance.

Corrected:

> On average, general election advertisements move candidate favorability by **0.123 scale points** more than primary advertisements, although the estimate is not precise enough to achieve statistical significance.

Table 1 of the same article prints 0.123 in this cell.

Table 1 also prints the model 1 battleground coefficient as −0.00, where the deposited code gives −0.008; the facing text quotes its magnitude correctly as 0.008. The table is reprinted below with that cell at the three decimals the rest of the column carries. Every other cell is unchanged.

The sentence carries one number downstream, three sentences later: the upper bound of the 95 per cent interval for the same difference, which the article gives as about 0.25. Computed from the corrected coefficient it is 0.254, so the bound and the reading built on it, that any heterogeneity by election phase is too small to matter politically, stand as published.

Meta-analysis of average treatment effects of advertisements on target candidate favorability and vote choice.

	Candidate favorability (1)	Candidate favorability (2)	Vote choice (3)	Vote choice (4)
Average effect	0.056*	0.062*	0.007	0.008
	(0.020)	(0.020)	(0.007)	(0.007)
Democratic respondent (versus Republican)	0.035	0.022	0.011	0.006
	(0.035)	(0.036)	(0.010)	(0.011)
Independent respondent (versus Republican)	0.023	0.015	0.009	0.007
	(0.051)	(0.052)	(0.020)	(0.020)
Battleground state (versus non-battleground)	-0.008	-0.007	-0.017	-0.017
	(0.033)	(0.033)	(0.010)	(0.010)
PAC sponsor (versus campaign sponsor)	-0.012	0.026	-0.023	-0.016
	(0.043)	(0.047)	(0.013)	(0.014)
Time (scaled in months)	-0.023	0.005	-0.009*	-0.008
	(0.014)	(0.010)	(0.004)	(0.004)
Attack advertisement (versus promotional advertisement)	-0.017		0.028	
	(0.046)		(0.016)	
General election (versus primary election)	0.123			
	(0.067)			
Pro-Trump advertisement (versus pro-Clinton advertisement)		-0.124		-0.016
		(0.101)		(0.034)
Anti-Clinton advertisement (versus pro-Clinton advertisement)		-0.105		0.012
		(0.070)		(0.023)
Anti-Trump advertisement (versus pro-Clinton advertisement)		-0.041		0.026
		(0.058)		(0.021)
Pro-Sanders advertisement (versus pro-Clinton advertisement)		-0.075		
		(0.089)		
Pro-Cruz advertisement (versus pro-Clinton advertisement)		0.047		
		(0.116)		
Pro-Kasich advertisement (versus pro-Clinton advertisement)		-0.182		
		(0.145)		
Number of observations	354	354	204	204

* $p < 0.05$

Note: Table 1 as published, with the model 1 battleground coefficient printed to three decimals. Standard errors in parentheses; $*$ denotes $p < 0.05$. Estimated from the deposited replication archive by `maintained/table_1_meta_regression.R`.

3. No positive estimate in the study exceeds half a scale point

Published, Results:

> …one large positive result toward the end of the campaign in October (**more than +0.5 points**), and two other negative results in October.

Corrected:

> …one large positive result toward the end of the campaign in October (**more than +0.4 points**), and two other negative results in October.

The largest positive weekly estimate anywhere in the study is 0.432, on the advertisement "Example" in the week of 31 October. The passage's point, that a significance filter applied to these experiments would leave a handful of results that look large, is unaffected.

The filter is what the corrected number is the warrant for, and it can be applied: of the 59 weekly favorability experiments, five have an interval excluding zero, and they run from -0.521 to 0.432. The passage names the four largest of those five in magnitude. So a filtered literature drawn from this study would show a spread of about a fifth of the favorability scale, which is the appearance of heterogeneity the passage is describing, and the corrected bound does not soften it. The title of this

entry says positive because the study's most extreme estimate of either sign is −0.521, the early May result the same sentence describes as nearly −0.5.

4. The advertisement counts in section A disagree with the main text

Published, app. Section A:

> Including the ads we deployed more than once, we tested **28** ads that attacked Trump, **nine** attacks on Clinton, **nine** Clinton promotional ads, and three promotional Trump ads.

Corrected:

> Including the ads we deployed more than once, we tested **30** ads that attacked Trump, **11** attacks on Clinton, **8** Clinton promotional ads, and three promotional Trump ads.

Counted as the section says, over advertisement-week deployments rather than distinct advertisements, the breakdown is 30, 11, 8 and 3, which is exactly what the main text states. Counted over distinct advertisements it is 24, 11, 6 and 3. The published breakdown matches neither.

The four counts are a partial inventory, and correcting them resizes what is left. They now sum to 52 of the 59 advertisement-week experiments, leaving 7 for the promotional advertisements for other primary candidates the next sentence calls the remainder, where the published counts left 10. Nothing estimated in the article is computed from this split: it is an inventory of the treatments, and its warrant is that a reader can reconcile section A with the main text, which the published version does not allow.

5. The number of unique advertisements in section A disagrees with the abstract

Published, app. Section A:

> Table S1 shows the name and target of the **50** unique advertisements and the week they were deployed.

Corrected:

> Table S1 shows the name and target of the **49** unique advertisements and the week they were deployed.

The deposited data hold 49 campaign advertisements plus the placebo, and the abstract's 49 is right. The abstract, the introduction and the discussion all say 49, so section A was the one place in the paper that said 50.

The sentence also defines Table S1, and the corrected number does not reach it. That table prints one cell per advertisement-week with a distinct title, which is 58 cells carrying 48 distinct names, so a reader who counts it recovers neither the 49 advertisements nor the 59 experiments. Two of the 49 advertisements share the title "Sacrifice", the target Trump and the week of 5 September, and the table has no way to show them apart. The shortfall is in the table's presentation and not in the study: both advertisements are in the deposited data and both are in every estimate the article reports.

6. The manipulation check range is one point narrow at each end

Published, Materials and Methods:

> Tiny fractions of the control groups claimed to have seen campaign advertisements (2 to 4%) compared with large majorities of the treatment groups (**92 to 95%**).

Corrected:

> Tiny fractions of the control groups claimed to have seen campaign advertisements (2 to 4%) compared with large majorities of the treatment groups (**91 to 96%**).

Across the ten waves in which the manipulation check was asked, the unweighted share of treated respondents reporting a campaign advertisement runs from 91.4 to 95.5 per cent, which is what the deposited script written for this sentence computes. Weighting by the survey weights widens the range further, to 90.2 to 95.9 per cent. The control group range is correct as published.

The sentence is the article's warrant that treatments were delivered as intended, and what carries that claim is the low end rather than the range. Correcting it moves the worst wave down by one point, to 91.4 per cent, which is still 23 times the highest rate any control group reported. Delivery is not in question in any wave.

7. Standard errors do not exceed the coefficients in every partisanship cell

Published, Results:

> Turning first to features of receiver, we see that the differences in treatment response across Democrats, Independents, and Republicans are small, **with SEs greater than coefficients**.

Corrected:

> Turning first to features of receiver, we see that the differences in treatment response across Democrats, Independents, and Republicans are small, **with SEs greater than coefficients in 6 of the 8 cells**.

The two exceptions are the Democratic respondent coefficients in columns 1 and 3 of Table 1, and the second is visible on the printed page: 0.011 with a standard error of 0.010. Neither coefficient is distinguishable from zero, so the claim the sentence supports, that treatment response does not differ by respondent partisanship, is unaffected.

The count is a proxy and the quantity behind it is the better warrant: across all 8 cells the largest ratio of a coefficient to its own standard error is 1.08, so no partisanship difference in the table reaches even 1.1 standard errors from zero. Six of eight is what the published sentence would have read as a weaker version of its own claim; the ratio says the claim is as strong as the sentence makes it.

8. Section A names eight advertisements shown in multiple weeks

Published, app. Section A:

> Some advertisements (**"America," "Love/Kindness," "Man of People," "Mirrors," "Quotes," "Real Life," "Role Model," "Speak"**) were shown in multiple weeks, enabling us to estimate how the effect of the same advertisement may vary with the changing electoral context.

Corrected:

> Some advertisements (**"America," "Love/Kindness," "Mirrors," "Quotes," "Real Life," "Role Model," "Speak"**) were shown in multiple weeks, enabling us to estimate how the effect of the same advertisement may vary with the changing electoral context.

"Man of People" was fielded once, in the week of 14 March. Section B and Figure S1 both use the correct count of 7.

The list is not a label on the sentence: it is the stated roster of what Figure S1 shows. That figure carries one panel per advertisement fielded in more than one week, and its seven panels are the corrected seven names exactly, in order. The published eighth name therefore promises a panel that is not in the figure, and section B, two pages later, already says seven. The analysis is unaffected because the figure and the tests behind it were always the correct seven: what was wrong was the only place a reader is told which advertisements they are.

9. "treatmen" for "treatment"

Published, Materials and Methods:

> OLS is a consistent estimator of the average **treatmen** effect (ATE) under our design (35).

Corrected:

> OLS is a consistent estimator of the average **treatment** effect (ATE) under our design (35).

A dropped letter, in the sentence that names the estimand. The article spells the term correctly everywhere else, including in the sentence immediately before this one. Found by the remaster, which carries the published text across word by word. No quantity is involved, so the question this note asks of the other eight entries, what the corrected value was the warrant for, does not arise. What the correction restores is that the sentence defining the estimand spells it, which is the one sentence in the article where the term is doing definitional work rather than referring back.